

Kelsee Bratley, PhD

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Education

Ph.D. in Earth and Environment 09/2019 – 08/2025

Boston University, Department of Earth and Environment

Boston, MA

Advisor: Dr. Curtis E. Woodcock

Dissertation Title: Remote sensing for agricultural assessments and ecological management interventions

M.A. in Remote Sensing and GIS 09/2018 – 05/2019

Boston University, Department of Earth and Environment

Boston, MA

B.Sc. in Environmental Science 08/2014 – 05/2018

University of North Carolina Wilmington, Department of Environmental Sciences

Wilmington, NC

Advisor: Dr. Eman Ghoneim

Honor's Thesis: Modeling Urban Encroachment on the Agricultural Land of the Eastern Nile Delta Using Remote Sensing and a GIS-Based Markov Chain Model

Work Experience

Doctoral Researcher 09/2019 – 08/2025

Boston University, Department of Earth and Environment

Boston, MA

- Designed and implemented custom GIS workflows and applications across diverse domains (agriculture, ecology, land use change), collaborating with government agencies and international organizations to solve complex spatial analysis challenges spanning multiple countries.

Research Contractor & Collaborator 03/2023 – 12/2024

International Monetary Fund

Maputo, Mozambique

- Developed advanced remote sensing solutions including machine learning classification to map agricultural extent across Mozambique, coordinating with UN agencies, government ministries, international geospatial organizations, and local universities.

Research Assistant 09/2018 – 08/2019

Boston University

Boston, MA

- Digitized agricultural field data using QGIS to support land-cover classification across diverse global environments.

Skills

Remote Sensing Methods: Change detection, land cover classification, area estimation, machine learning (Random Forest, SVM, Gradient Boosting, K-means clustering), design- and model-based statistical inference, fixed-effects regression, big data management

Programming: Google Earth Engine, R, Python, GDAL, High performance computing (Linux)

Applications: ArcGIS, QGIS, ENVI, Microsoft Suite

Teaching Experience

Teaching Fellow, Boston University

Spring 2020 & Spring 2025

Course: Digital Image Processing (Graduate Level)

- Taught advanced image processing techniques including image classification, change detection, map accuracy and area estimation, and object-based segmentation. Used Python, Bash, and Google Earth Engine in hands-on labs, while mentoring students on independent project work.

Instructor of Record, Boston University

Fall 2024

Course: Remote Sensing of the Environment

- Designed and managed all aspects of an undergraduate Remote Sensing course, including developing curriculum, delivering lectures, facilitating class discussions, and providing detailed feedback to enhance student understanding of remote sensing principles and applications.

Teaching Fellow, Boston University

Fall 2019

Course: Geographic Information Systems (GIS) (Graduate Level)

- Taught GIS lab sessions focused on spatial analysis, geoprocessing, and cartographic visualization using tools such as ArcGIS and QGIS.

Teaching Fellow, Boston University

Spring 2019

Course: Introduction to Climate and Earth System Science

Teaching Fellow, Boston University

Fall 2018

Course: Biology 1

Publications

Peer-Reviewed

Bratley, K. H. & Woodcock, C. E. (2024). Estimating the expansion and reduction of agricultural extent in Egypt using Landsat time series. *International Journal of Applied Earth Observation and Geoinformation*, 133, 104141. <https://doi.org/10.1016/j.jag.2024.104141>

Tang, X., Barrett, M. G., Cho, K., Bratley, K. H., Tarrio, K., Zhang, Y., Gu, H., Rasmussen, P., Bosch, M., & Woodcock, C. E. (2024). Broad-area-search of new construction using time series analysis of Landsat and Sentinel-2 data. *Science of Remote Sensing*, 9, 100138. <https://doi.org/10.1016/j.srs.2024.100138>

Tang, X., Bratley, K. H., Cho, K., Bullock, E. L., Olofsson, P., & Woodcock, C. E. (2023). Near real-time monitoring of tropical forest disturbance by fusion of Landsat, Sentinel-2, and Sentinel-1 data. *Remote Sensing of Environment*, 294, 113626. <https://doi.org/10.1016/j.rse.2023.113626>

Pasquarella, V. J., Arévalo, P., Bratley, K. H., Bullock, E. L., Gorelick, N., Yang, Z., & Kennedy, R. E. (2022). Demystifying LandTrendr and CCDC temporal segmentation. *International Journal of Applied Earth Observation and Geoinformation*, 110, 102806. <https://doi.org/10.1016/j.jag.2022.102806>

Bratley, K. H. & Ghoneim, E. (2018). d. *Land (Basel)*, 7(4), 114. <https://doi.org/10.3390/land7040114>

Submitted (Under Review)

Dong, L., Tang, X., Tarrio, K., Bratley, K.H., Mensah, F., Abubakari, B., Amponsah, M., Olofsson, P., & Woodcock, C. (2025). Satellite analysis reveals loss of native forest in Ghana is high, but lower than prior indicators. Manuscript submitted to *Global Change Biology*.

In-Preparation

Bratley, K. H., & Woodcock, C. E. (2025). Mapping agricultural extent in Africa: Progress, past challenges, and future opportunities. In preparation.

Bratley, K. H., Kaufmann, R., Foley, F., Grissom, P., Nolte, C., Cho, K., Woodcock, C. E. (2025). Evaluating Treatment Efficacy on Invasive Buffelgrass (*Cenchrus ciliaris*) in Saguaro National Park Using Landsat Time-Series and Fixed Effects Regression Modeling. In preparation.

Non-Peer-Reviewed

Bratley, K. H., & Meyer-Cirkel, A. (2025). Understanding Agricultural Output in Mozambique: Using Remote Sensing to Initiate a Discussion on Development. White Paper, International Monetary Fund.

Conference Presentations

Bratley, K. H., Nolte, C., Kaufmann, R., Cho, K., & Woodcock, C. E. (2024, December). Evaluating treatment efficacy on invasive buffelgrass (*Cenchrus ciliaris*) in Saguaro National Park using Landsat time-series and fixed effects panel regression [Poster presentation]. AGU Fall Meeting, Washington, D.C.

Bratley, K. H., & Woodcock, C. E. (2022, December). Mapping multiple kinds of agricultural land-cover change in Egypt using a LandTrendr time series temporal segmentation approach [Poster presentation]. AGU Fall Meeting, Chicago, IL.

Bratley, K. H., & Woodcock, C. E. (2022, October). Mapping multiple kinds of agricultural land-cover change in Egypt using a LandTrendr time series temporal segmentation approach [Oral presentation]. Pecora Conference, Denver, CO.

Bratley, K. H., & Ghoneim, E. (2019, March). Modeling urban encroachment on the agricultural land of the Eastern Nile Delta using remote sensing and a GIS-based Markov chain model [Poster presentation]. International Scientific Research Conference on Renewable Energy and Water Sustainability, Sharm El-Sheikh, Egypt.

Bratley, K. H., & Ghoneim, E. (2018, April). Urban encroachment on the Nile Delta [Poster presentation]. Association of American Geographers Annual Meeting, New Orleans, LA.

Invited Lectures & Talks

Bratley, K. H. (2024, June). Mapping Agricultural Extent in Mozambique [Invited speaker]. Digital Earth Africa.

Bratley, K. H. (2020–2025, Spring). Quantifying Spatial and Temporal Agricultural Trends: A Guide to Using LandTrendr [Guest lecture]. Boston University, Digital Image Processing.

Bratley, K. H. (2022, Fall). Assessing Multiple Kinds of Agricultural Land-Cover Change in Egypt Using LandTrendr [Guest lecture]. James Madison University, Introduction to Remote Sensing.